

#1a: nuclear timescale

(i) 2 points: 1 pt for attempt; 1 pt for correct; -0.5 pt for no units

0.7 hydrogen mass fraction; $f_{\text{nuc}} = 0.13$

$$\begin{aligned}\rightarrow M_{\text{nuc}} &= (0.7)(0.13) (1 M_{\odot}) \\ &= 0.091 M_{\odot} \\ &= (0.091)(1.989 \times 10^{30} \text{ kg}) \\ &= 1.8 \times 10^{29} \text{ kg}\end{aligned}$$

(ii) 2 points: 1 pt for attempt; 1 pt for correct

$$\tau_{\text{nuc}} = \frac{E_{\text{nuc}}}{L} = \frac{\phi M_{\text{nuc}} c^2}{L}$$

$$\phi = 0.007, \quad M_{\text{nuc}} = 0.091 M$$

$$\begin{aligned}\rightarrow \tau_{\text{nuc}} &= \frac{\phi c^2 (0.091 M)}{L} \left(\frac{3.826 \times 10^{26} \frac{\text{kg m}^2}{\text{s}^3}}{L_{\odot}} \right) \left(\frac{1.989 \times 10^{30} \text{ kg}}{M_{\odot}} \right) \\ &= 2.98 \times 10^{17} \text{ s} \left(\frac{M}{M_{\odot}} \right) \left(\frac{L_{\odot}}{L} \right)\end{aligned}$$

$$1 \text{ yr} = 3.155 \times 10^7 \text{ s}, \text{ so}$$

$$\tau_{\text{nuc}} = 9.4 \times 10^9 \text{ yr} \left(\frac{M}{M_{\odot}} \right) \left(\frac{L_{\odot}}{L} \right) \approx 10^{10} \text{ yr}$$

(iii) 2 points: 1pt for attempt, 1pt for correct.

$$M-L \text{ relation: } \frac{L}{L_{\odot}} = \left(\frac{M}{M_{\odot}} \right)^{3.8}$$

$$\begin{aligned} \rightarrow \tau_{\text{nuc}} &\approx 10^{10} \text{ yr} \left(\frac{M}{M_{\odot}} \right) \frac{1}{\left(\frac{M}{M_{\odot}} \right)^{3.8}} \\ &\approx 10^{10} \text{ yr} \left(\frac{M}{M_{\odot}} \right)^{-2.8} \end{aligned}$$

(iv) The nuclear timescale describes how long a star will take to burn through its fuel.
(2pts for correct interpretation)

#1b the thermal timescale

(i) 2 points: 1pt for attempt, 1pt for correct

$$\tau_{\text{thermal}} = \tau_{\text{KH}} = \frac{GM^2}{2RL}$$

$$\begin{aligned} \tau_{\text{KH}} &= \frac{6.67 \times 10^{-11} \frac{\text{m}^3}{\text{kg s}^2} \frac{\text{M}^2}{\text{RL}} \left(\frac{1.989 \times 10^{30} \text{ kg}}{M_{\odot}} \right)^2 \left(\frac{R_{\odot}}{6.957 \times 10^8 \text{ m}} \right) \left(\frac{L_{\odot}}{3.826 \times 10^{26} \frac{\text{kg m}^2}{\text{s}^3}} \right)}{2} \\ &= 4.96 \times 10^4 \text{ s} \left(\frac{M}{M_{\odot}} \right)^2 \left(\frac{R_{\odot}}{R} \right) \left(\frac{L_{\odot}}{L} \right) \end{aligned}$$

$$\rightarrow \tau_{\text{KH}} = 1.5 \times 10^7 \text{ yr} \left(\frac{M}{M_{\odot}} \right)^2 \left(\frac{R_{\odot}}{R} \right) \left(\frac{L_{\odot}}{L} \right)$$

(ii) $\frac{L}{L_\odot} = \left(\frac{M}{M_\odot}\right)^{3.8}$; $\frac{R}{R_\odot} = \left(\frac{M}{M_\odot}\right)^{0.7}$ 2 points: 1pt attempt
1 pt correct

$$\begin{aligned}\tau_{\text{KH}} &= 1.5 \times 10^7 \text{ yr} \left(\frac{M}{M_\odot}\right)^2 \left(\frac{R_\odot}{R}\right) \left(\frac{L_\odot}{L}\right) \\ &= 1.5 \times 10^7 \text{ yr} \left(\frac{M}{M_\odot}\right)^2 \frac{(2 - 0.7 - 3.8)}{\left(\frac{M}{M_\odot}\right)^{0.7} \left(\frac{M}{M_\odot}\right)^{3.8}} = 1.5 \times 10^7 \text{ yr} \left(\frac{M}{M_\odot}\right)^{-2.5} \\ &= 1.5 \times 10^7 \left(\frac{M}{M_\odot}\right)^{-2.5} = \tau_{\text{KH}}\end{aligned}$$

(iii) The thermal timescale sets how long it takes a star to regain thermal equilibrium after being Perturbed.

2 points

#1C dynamical timescale

$$\begin{aligned}(i) \tau_{\text{dyn}} &= \sqrt{\frac{R^3}{GM}} = \left(6.67 \times 10^{-11} \frac{\text{m}^3}{\text{kg s}^2}\right)^{-1/2} R^{3/2} M^{-1/2} \\ &= 1.22 \times 10^5 \frac{\text{kg}^{1/2} \text{s}}{\text{m}^{3/2}} \frac{R^{3/2}}{M^{1/2}} \left(\frac{M_\odot}{1.989 \times 10^{30} \text{ kg}}\right)^{1/2} \left(\frac{6.957 \times 10^8 \text{ m}}{R_\odot}\right)^{3/2} \\ &= 1587 \text{ s} \left(\frac{R}{R_\odot}\right)^{3/2} \left(\frac{M_\odot}{M}\right)^{1/2} \\ &\approx 0.02 \text{ day} \left(\frac{R}{R_\odot}\right)^{3/2} \left(\frac{M_\odot}{M}\right)^{1/2}\end{aligned}$$

2 points: 1pt for attempt; 1pt for correct

(ii) 2 points: 1 pt for attempt; 1 pt for correct

$$\tau_{\text{dyn}} \approx 0.02 \text{ day} \left(\frac{R}{R_{\odot}} \right)^{3/2} \left(\frac{M_{\odot}}{M} \right)^{1/2}$$

$$\frac{R}{R_{\odot}} = \left(\frac{M}{M_{\odot}} \right)^{0.7} \rightarrow \left(\frac{R}{R_{\odot}} \right)^{3/2} = \left[\left(\frac{M}{M_{\odot}} \right)^{0.7} \right]^{3/2} \\ = \left(\frac{M}{M_{\odot}} \right)^{1.05}$$

$$\tau_{\text{dyn}} \approx 0.02 \text{ day} \left(\frac{M}{M_{\odot}} \right)^{1.05} \left(\frac{M_{\odot}}{M} \right)^{1/2} \\ \approx 0.02 \text{ day} \left(\frac{M}{M_{\odot}} \right)^{0.55}$$

(iii) 2 pts

The dynamical timescale is the time it takes a star to react to perturbations of hydrostatic equilibrium.

It is also how fast a star would collapse under gravity if the pressure was suddenly removed.

#1d comparing the timescales

(i) 3 points: 1.5 for attempt; 0.5 each for correct

$$\tau_{\text{nuc}} \approx 10^{10} \text{ yr} \left(\frac{M}{M_{\odot}} \right)^{-2.8}$$

$$\rightarrow \tau_{\text{nuc}, 1M_{\odot}} \approx 10^{10} \text{ yr}$$

$$\tau_{\text{nuc}, 10M_{\odot}} \approx 10^{10} \text{ yr} (10)^{-2.8} \approx 1.585 \times 10^7 \text{ yr}$$

$$\tau_{\text{nuc}, 25M_{\odot}} \approx 10^{10} \text{ yr} (25)^{-2.8} \approx 1.212 \times 10^6 \text{ yr}$$

(i) continued

$$\tau_{KH} \approx 1.5 \times 10^7 \text{ yr} \left(\frac{M}{M_\odot} \right)^{-2.5}$$

$$\rightarrow \tau_{KH, 1M_\odot} \approx 1.5 \times 10^7 \text{ yr}$$

$$\tau_{KH, 10M_\odot} \approx 1.5 \times 10^7 \text{ yr} (10)^{-2.5} \approx 4.74 \times 10^4 \text{ yr}$$

$$\tau_{KH, 25M_\odot} \approx 1.5 \times 10^7 \text{ yr} (25)^{-2.5} \approx 4800 \text{ yr}$$

$$\tau_{dyn} \approx 0.02 \text{ day} \left(\frac{M}{M_\odot} \right)^{0.55}$$

$$\rightarrow \tau_{dyn, 1M_\odot} \approx 0.02 \text{ day}$$

$$\tau_{dyn, 10M_\odot} \approx 0.02 \text{ day} (10)^{0.55} = 0.071 \text{ day} \approx 1.7 \text{ hr}$$

$$\tau_{dyn, 25M_\odot} \approx 0.02 \text{ day} (25)^{0.55} = 0.12 \text{ day} \approx 2.8 \text{ hr}$$

(ii) 4 points: 1pt each for attempt, 1pt each for correct

$$\tau_{dyn} \approx 0.02 \text{ day} \left(\frac{R}{R_\odot} \right)^{3/2} \left(\frac{M_\odot}{M} \right)^{1/2} \quad \tau_{KH} = 1.5 \times 10^7 \text{ yr} \left(\frac{M}{M_\odot} \right)^2 \left(\frac{R_\odot}{R} \right) \left(\frac{10}{L} \right)$$

$$R = 200 R_\odot, M = 1 M_\odot \rightarrow \tau_{dyn} \approx 0.02 \text{ day} (200)^{3/2} (1)^{1/2} \\ \approx 56.6 \text{ day}$$

$$\rightarrow \tau_{KH} \approx 1.5 \times 10^7 \text{ yr} (1)^2 \left(\frac{1}{200} \right) \left(\frac{1}{500} \right) \\ \approx 25 \text{ yr}$$

(iii) 2 points; 1 for attempt, 1 for correct

$$\tau_{\text{dyn}} \geq \tau_{\text{KH}} \rightarrow \sqrt{\frac{R^3}{GM}} \geq \frac{GM^2}{2RL}$$

solve for R :

$$\frac{R^3}{GM} \geq \frac{G^2 M^4}{4R^2 L^2} \rightarrow R^5 \geq \frac{G^3 M^5}{4 L^2}$$

check units:

$$\left[\frac{\text{m}}{\text{kg s}^2} \right]^3 [\text{kg}]^5 \left[\frac{\text{s}^3}{\text{kg m}^2} \right]^2 = \left[\frac{\text{m}^9}{\text{kg}^3 \text{s}^6} \right] [\text{kg}^5] \left[\frac{\text{s}^6}{\text{kg}^2 \text{m}^4} \right]$$
$$= [\text{m}^5] \text{ yay!}$$

use: $L = 4\pi\sigma R^2 T^4$:

$$R^5 \geq \frac{G^3 M^5}{4 (4\pi\sigma^2 T^8 R^4)}$$

$$\rightarrow R^9 \geq \frac{G^3 M^5}{64\pi^2 \sigma^2 T^8}$$

$$M = 1.989 \times 10^{30} \text{ kg}$$

$$G = 6.67 \times 10^{-11} \text{ m}^3/\text{s}^2\text{kg}$$

$$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$$

$$T = 5770 \text{ K}$$

$$\rightarrow R \geq \left[\frac{G^3 M^5}{64\pi^2 \sigma^2 T^8} \right]^{1/9} = 2.5 \times 10^8 \text{ m} = 360 R_{\odot}$$

$$\rightarrow R \geq 360 R_{\odot}$$

$$\text{check: } \tau_{\text{dyn}} = \sqrt{R^3/GM} \approx 1 \times 10^7 \text{ s} \approx \frac{1}{3} \text{ yr}$$

$$\tau_{\text{KH}} = \frac{GM^2}{2RL} \approx 1 \times 10^7 \text{ s} \approx \frac{1}{3} \text{ yr}$$

Problem 2 : effectively participation points

(2a) 3 points, 1 each for search term, #, answer

(2b) 5 points, 1 each for each search term } #
1 for answer

(2c) 5 points, 2 per paper selection, 1 for answer.